

Final Project Guide

Scaler School of Technology | Deep Learning I: Neural Networks | 35 Marks



Project Overview

The Final Project is the capstone evaluation for this course. It combines a **group project**, a **group presentation**, and an **individual viva** into a single 35-mark component. Each group of **8 students** (assigned randomly) works on **two tasks**: one end-to-end applied project and one focused conceptual/experimental task.

Two Tasks Per Group

Task 1 (Section 1): A small, interesting end-to-end project — build something real, train it, evaluate it. This builds **confidence** and hands-on skills.

Task 2 (Section 2): A focused experimental task designed to build **theoretical and conceptual foundations** — understand *why* things work, not just *how* to make them run.

Marks Breakdown (35 Marks)

Component	Marks	Weight	Scope
Project + Presentation (Technical)	~21	60%	Group
Presentation Skills	~4	10%	Group
Individual Viva	~11	30%	Individual
Total	35	100%	

- **70% (~25 marks)** is based on the project deliverables and group presentation
- **30% (~11 marks)** is based on the individual viva of **one randomly chosen member**
- The viva candidate is announced **one day before** the presentation — *so everyone must prepare*
- The viva score of that one member becomes the viva score for the **entire group**

Group Formation & Contribution Policy

- Groups of **8 students**, assigned **randomly**
- Each group picks (or is assigned) **one topic from Section 1** and **one topic from Section 2**

Vote-Out Policy for Non-Contributing Members

If a group member is **not making any contributions**, the rest of the group can **vote them out** (requires majority vote + evidence submitted to the TA).

For voted-out students:

- No penalty for leaving the group
- They must complete the **same topic** as an **individual project**
- Same deadlines, same rubric — just solo
- This is meant as a safety net, not a punishment — but the workload is real

Why Random Viva Selection?

Since one person's viva determines 30% of the group's marks, **every member** has an incentive to ensure that **every other member** understands the project deeply. This creates natural peer teaching and shared accountability. The announcement is made one day before so the group can do a final prep session together.

Section 1 — End-to-End Applied Projects (Build Confidence)

Each group picks **one** topic from this section. These are complete, hands-on projects where you build something real from data to evaluation. Focus: *can you build it, train it, and make it work?*

1 Image Classification: Architecture Comparison

Classify images on CIFAR-10. Compare a simple CNN (2–3 conv layers), a deeper CNN with batch normalization, and the deeper CNN trained with data augmentation (random crop, horizontal flip). Analyze accuracy, training time, parameter count, and loss curves.

2 Sentiment Analysis with Sequence Models

Binary sentiment classification on IMDB. Compare an MLP baseline on TF-IDF features, a vanilla RNN, and an LSTM. Analyze accuracy, training stability, and how well each model handles long reviews.

3 Transfer Learning on a Small Dataset

Pick a small image dataset (500–2000 images, e.g. from Kaggle). Compare training a CNN from scratch vs. using a pre-trained ResNet-18 with frozen backbone vs. fine-tuning the last layer. Show when transfer learning helps.

4 Sequence-to-Sequence for Date Conversion

Build a seq2seq model to convert date formats (e.g. “January 5, 2023” → “2023-01-05”). Compare a basic encoder-decoder vs. one with attention. Visualize attention weights on sample inputs. Report accuracy and convergence speed.

5 Training Dynamics: Optimizer & Regularization Study

Pick one dataset and architecture. Compare SGD vs. Adam, and no regularization vs. dropout vs. weight decay. Plot loss curves, validation accuracy, and training time for each combination. Explain which setup works best and why.

6 Time Series Forecasting

Forecast a real-world time series (weather or energy data). Compare a simple moving average baseline, a vanilla RNN, and an LSTM. Report MAE/RMSE, plot forecasts, and analyze how the lookback window length affects predictions.

7 Multi-Label Image Tagging

Use a dataset where images have multiple tags (e.g. Pascal VOC subset). Build a CNN with sigmoid outputs (not softmax) and compare with a deeper CNN with batch norm. Evaluate using precision, recall, and qualitative failure analysis.

8 Text Generation with Character-Level RNN

Train a character-level LSTM on a text corpus (e.g. Shakespeare). Generate text by sampling at different temperatures (0.5, 1.0, 1.5). Compare output quality at different training stages (early, mid, late). Analyze coherence and creativity.

9 Movie Review Star Predictor

Predict star ratings (1–5) from movie reviews. Compare an MLP on TF-IDF features vs. an LSTM on raw text. Analyze where models confuse adjacent ratings (3 vs. 4) and plot a confusion matrix for each approach.

10 Music/Audio Genre Classification

Classify audio clips by genre using mel-spectrograms as input to a CNN. Compare a simple CNN vs. a deeper CNN with batch norm. Evaluate per-genre accuracy, plot a confusion matrix, and analyze which genres are hardest to distinguish.

11 Fake News Detector

Build a fake news classifier on the LIAR dataset. Compare an MLP on hand-crafted text features vs. an LSTM on the article text. Evaluate precision and recall, and examine examples the model gets wrong.

12 Document Classification

Classify news articles (e.g. AG News) into categories. Compare an MLP on TF-IDF features vs. an LSTM. Analyze accuracy, training time, and which categories are easiest/hardest to classify.

13 Image Captioning with CNN + LSTM

Build an image captioning pipeline: use a pre-trained CNN (e.g. ResNet-18) as the encoder and an LSTM as the decoder. Train on Flickr8k. Show sample captions, compare with ground truth, and evaluate with BLEU scores.

14 Simple Machine Translation

Build an RNN-based seq2seq model with attention for a small translation task (e.g. Multi30k EN-DE). Report BLEU scores, show sample translations, and visualize attention alignment for a few examples.

15 Text Summarization with Seq2Seq

Build an extractive or simple abstractive summarizer using seq2seq with attention on a small subset of CNN/DailyMail. Compare with a lead-sentence baseline. Evaluate with ROUGE scores and show qualitative examples.

16 Medical Image Classification

Apply a pre-trained CNN to a medical imaging dataset (e.g. chest X-rays or skin lesions). Compare frozen backbone vs. fine-tuned backbone. Evaluate with accuracy and a confusion matrix. Discuss class imbalance challenges.

17 Emoji Prediction from Text

Given a tweet, predict the most likely emoji using an MLP on bag-of-words vs. an LSTM. Evaluate top-1 and top-3 accuracy. Analyze which emojis are easiest and hardest to predict and why.

18 Image Segmentation with a Simple FCN

Build a simple fully convolutional network (FCN) for pixel-wise segmentation on a small dataset (e.g. Oxford Pets). Compare with a baseline that predicts the most common class per pixel. Evaluate with pixel accuracy and visualize segmentation masks.

19 Spam/Toxic Comment Detection

Build a text classifier to detect toxic comments (e.g. Jigsaw dataset). Compare an MLP on TF-IDF vs. an LSTM. Analyze precision-recall trade-offs and examine misclassified examples to understand model failures.

20 Machine Translation: RNN Encoder-Decoder

Build an RNN encoder-decoder for a small translation task. Compare with and without attention. Report BLEU scores, show example translations, and analyze where the model struggles (long sentences, rare words).

21 AI Story Continuation

Train a character-level or word-level LSTM on a story corpus. Generate continuations from prompts using greedy vs. temperature-based sampling. Compare output quality and analyze how corpus style affects the generated text.

22 Paraphrase Detection

Train an LSTM-based model to detect whether two sentences are paraphrases (e.g. MRPC dataset). Compare with a cosine similarity baseline on TF-IDF vectors. Report accuracy and F1 score. Analyze failure cases.

23 Food Image Classifier

Classify food images into categories (e.g. Food-101 subset with 10 classes) using a CNN. Compare training from scratch vs. fine-tuning a pre-trained model. Analyze which food categories are most confused and why.

24 Single-Task vs. Multi-Output CNN

Train a CNN on an image dataset with two labels (e.g. object class + coarse category). Compare a single-output model trained twice vs. a shared-backbone model with two output heads. Analyze whether sharing helps or hurts.

25 Meme Sentiment Classifier

Classify memes as positive or negative using image features (CNN) and caption text (LSTM) separately. Compare image-only vs. text-only accuracy. Optionally try a simple late fusion (concatenate features). Analyze where each modality fails.

26 News Article Categorization: Classical vs. Deep

Classify news articles (e.g. AG News). Compare TF-IDF + logistic regression vs. an MLP on bag-of-words vs. an LSTM. Analyze accuracy and training cost. Discuss when deep learning provides meaningful gains over classical methods.

Section 2 — Conceptual & Experimental Tasks (Build Foundations)

Each group picks **one** topic from this section. These are focused, controlled experiments designed to deepen your **theoretical and conceptual understanding**. Focus: *why does this work? what happens when you change X?*

1 Optimizer Showdown

Train an MLP on Fashion-MNIST with SGD, SGD+Momentum, and Adam. Compare convergence speed, final accuracy, and loss curves. Explain *why* Adam converges faster than vanilla SGD.

2 Regularization Effects on Overfitting

Train an MLP on a **small subset** of a dataset (to induce overfitting). Compare: no regularization, dropout, and weight decay. Measure train-val accuracy gap. Show which technique reduces overfitting most.

3 CNN Filter Visualization

Train a CNN on CIFAR-10. Extract and plot first-layer filters. Visualize feature maps at an early and a late layer. Show how representations become more abstract with depth.

4 How Much Training Data Do You Need?

Train the same CNN on CIFAR-10 using 5%, 25%, 50%, and 100% of the data. Plot accuracy vs. dataset size. Add data augmentation and repeat. Show where augmentation helps the most.

5 Initialization Matters

Train a deep MLP (6+ layers) with zero init, random normal, Xavier, and He initialization. Show which ones allow training and which fail. Plot loss curves for each.

6 Transfer Learning: Frozen vs. Fine-Tuned

Use a pre-trained ResNet-18 on a small image dataset. Compare: frozen backbone (feature extraction) vs. fine-tuning the last layer vs. training from scratch. Plot accuracy for each approach.

7 Activation Function Comparison

Train the same MLP with Sigmoid, ReLU, and Leaky ReLU. Plot loss curves and final accuracy. Show the vanishing gradient problem with sigmoid in a deeper network (6+ layers).

8 Learning Rate Schedule Comparison

Train a CNN with constant LR, step decay, and cosine annealing. Compare convergence speed and final accuracy. Plot the learning rate and loss curves side by side.

9 Batch Size Effects on Training

Train the same model with batch sizes 8, 64, and 512. Measure training speed, convergence, and final accuracy. Explain the relationship between batch size and learning rate.

10 Dropout Rate Sensitivity

Train the same network with dropout rates 0, 0.2, 0.5, and 0.7. Plot train/val accuracy for each. Show the sweet spot where dropout helps without hurting too much.

11 The Dying ReLU Problem

Create a setting where ReLU neurons die (e.g. high learning rate). Monitor the fraction of dead neurons. Compare ReLU vs. Leaky ReLU. Show which is more robust.

12 Data Augmentation Ablation

Train a CNN on a small image dataset. Systematically test: no augmentation, horizontal flip only, random crop only, and both together. Measure the impact on validation accuracy.

13 Loss Function Comparison

Train the same model with cross-entropy vs. cross-entropy with label smoothing on a balanced dataset. Compare accuracy and calibration (predicted probabilities vs. actual accuracy).

14 Can a Network Memorize Random Labels?

Train a CNN on CIFAR-10 with correct labels vs. fully random labels. Show that the network *can* memorize random data but only generalizes with real labels. Compare training loss curves.

15 Pooling Strategy Comparison

Train a CNN with max pooling vs. average pooling vs. global average pooling. Compare accuracy and parameter counts. Analyze which pooling works best for your dataset.

16 RNN vs. LSTM: Gradient Flow

Train a vanilla RNN and an LSTM on sequences of varying length (short, medium, long). Show that RNN performance degrades on longer sequences while LSTM remains stable. Plot training loss curves.

17 Shallow vs. Deep Networks

Train MLPs with 2, 4, and 8 hidden layers on Fashion-MNIST (same total parameter count). Plot accuracy for each depth. Show when deeper networks help and when they fail to train.

18 CNN Depth Analysis

Build CNNs with 3, 6, and 9 conv layers on CIFAR-10. Compare accuracy and training time. Show when adding more layers stops helping (diminishing returns).

19 How Many Parameters Does Your Model Really Need?

Train a CNN, then prune weights by magnitude (remove the smallest 30% and 50%). Measure accuracy after pruning. Show how many parameters can be removed without significant accuracy loss.

20 Bidirectional vs. Unidirectional LSTM

Compare a unidirectional LSTM vs. a bidirectional LSTM on a text classification task. Show when backward context helps. Report accuracy for each model.

21 Sequence Length Sensitivity in LSTMs

Train an LSTM on sequences of length 10, 50, and 200. Measure accuracy and training time. Show how performance changes with sequence length.

22 Fine-Tuning Depth: How Many Layers to Unfreeze?

Take a pre-trained CNN and fine-tune on a small dataset. Compare unfreezing 0, 1, and all layers. Plot accuracy for each. Show the best strategy for a small dataset.

23 Batch Norm vs. No Batch Norm

Train the same deep network (6+ layers) with and without batch normalization. Compare convergence speed, final accuracy, and training stability. Show how batch norm helps deep networks train.

24 Attention Head Count Study

Implement a simple Transformer block and vary the number of attention heads (1, 4, 8). Compare accuracy and training speed on a classification task. Discuss whether more heads always help.

25 Knowledge Distillation: Large → Small

Train a large CNN (teacher) and a small CNN (student). Compare: student trained alone vs. student trained with soft labels from the teacher. Show when distillation improves the student's accuracy.

26 Gradient Clipping in Deep Networks

Train a deep RNN with and without gradient clipping. Monitor gradient norms during training. Show how clipping stabilizes training and prevents exploding gradients.

Evaluation Guidelines & Metrics

Criterion	Marks	What We Look For
Project & Presentation — Technical (60% ≈ 21 marks, Group Score)		
Problem Understanding	~3	Clear problem statement, correct dataset choice, awareness of challenges
Implementation Quality	~6	Clean code, correct model implementation, proper data pipeline, reproducible notebook
Experiments & Results	~6	3+ meaningful experiments, controlled comparisons, clear plots/tables, statistical rigor
Analysis & Insights	~4	Why did X work? Why did Y fail? Conceptual depth, not just accuracy numbers
Report Quality	~2	Follows template, well-structured, clear figures with labels/legends
Presentation Skills (10% ≈ 4 marks, Group Score)		
Delivery & Communication	~2	Clear speech, good pacing, all members participate, within time limit
Visual Quality & Demo	~2	Clean slides, live demo or code walkthrough, effective visualizations
Individual Viva (30% ≈ 11 marks, Individual → Group)		
Project Understanding	~5	Can explain their group's approach, decisions, and results in depth
Conceptual Understanding	~6	Can answer theory questions from Weeks 1–7, connects theory to practice
Total	35	

Viva Format

- **One person** from each group is chosen **randomly**, announced **one day before**
- The viva lasts **8–10 minutes**: project questions + theory questions
- The evaluator picks from the theory question bank (shared with students in advance)
- The viva score of this one member becomes the **viva score for the entire group**
- This means: if your group's representative scores 4/11, **everyone** in the group gets 4/11

Viva Scoring Rubric

9–11: Explains confidently, handles follow-ups, connects theory to their project decisions

6–8: Good understanding, minor gaps on follow-up questions

3–5: Surface-level understanding, cannot explain why certain decisions were made

0–2: Cannot explain the project or answer basic theory questions

Student Deliverables (Due Day Before Presentation)

#	Deliverable	Format	Notes
1	Code Notebook (Task 1)	.ipynb with saved outputs	Must run top-to-bottom. Include all experiments and plots.
2	Code Notebook (Task 2)	.ipynb with saved outputs	Must run top-to-bottom. Focus on controlled experiments.
3	Project Report	PDF, 5–6 pages	Covers both tasks. Structure: Problem, Approach, Experiments, Analysis, Contribution.
4	Slide Deck	PDF or Google Slides (max 15 slides)	Cover both tasks. Every member presents at least 1 slide.

Group Scoresheet

Group #: _____ Section 1 Topic: _____ Section 2 Topic: _____

Panel: _____ Date: _____ Viva Candidate: _____

Part A: Project & Presentation — Technical (~21 marks)

Criterion	Max	Score
Problem Understanding	~3	_____
Implementation Quality	~6	_____
Experiments & Results	~6	_____
Analysis & Insights	~4	_____
Report Quality	~2	_____
Subtotal	~21	_____

Part B: Presentation Skills (~4 marks)

Criterion	Max	Score
Delivery & Communication	~2	_____
Visual Quality & Demo	~2	_____
Subtotal	~4	_____

Part C: Individual Viva (~11 marks) — Candidate: _____

Criterion	Max	Score
Project Understanding	~5	_____
Conceptual/Theory Understanding	~6	_____
Subtotal	~11	_____

Final Score

Technical (Part A)	~21	_____
Presentation Skills (Part B)	~4	_____
Viva (Part C)	~11	_____
TOTAL	35	_____

Group Members:

1. _____ 2. _____ 3. _____ 4. _____
 5. _____ 6. _____ 7. _____ 8. _____

Vote-out filed? ☐ No ☐ Yes (name: _____)

Evaluator signature: _____